PhotonicsAI — AI-Driven Science and Autonomous Laboratories

Homework 3: Hardware in the Loop

Name: Erik Mirzoyan

The device

A lab camera: the **Hikrobot MV-CS028-10UM** industrial camera (monochrome, USB 3.0, USB3 Vision), controlled through Hikrobot's MVS software development kit (SDK).

Project description

Until now the agent system from Homeworks 1 and 2 has worked only with code and with a simulated optical bench. Connecting the camera gives it a real instrument. The camera's controls (exposure, gain, region of interest, frame capture and image measurements) become tools on an MCP server, and one agent, the executor, holds them. A new hardware turn in the graph closes the loop: the agent decides what to do, commands the camera, reads the measurement back, and decides again. The human stop from Homework 2 stays for setting changes and for actions only a person can do, such as blocking the light.

With this connection the system will be able to carry out a measurement task from a plain-language request, choosing the settings itself and checking the result on the real image. It will set the exposure automatically so that the image uses most of the camera's range without saturating; characterise the camera's noise (read noise, conversion gain, full-well capacity) from dark frames and a photon-transfer series; find an exposure free of the room lights' flicker (mains lamps flicker at 100 Hz); and check the stability of the intensity over time. The lab has no laser for this homework, so the camera looks at a white surface under the room lights. The measured camera parameters will then be put into the camera model of the Homework 2 simulated bench, so that the agent predicts the focusing enhancement a wavefront-shaping experiment would reach with this camera: a first step towards running the Homework 2 focusing experiment on real hardware, with the lab's DMDs.